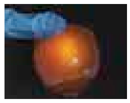
Eating fish may prevent asthma
Scientist claims an innovative study has revealed new evidence that eating fish can help prevent asthma. **https://dgit.in/apr19-3**



Self-healing, stretchable e-skin
Researchers have developed a touch-skin that could heal and help humans interact with machines. **https://dgit.in/apr19-4**

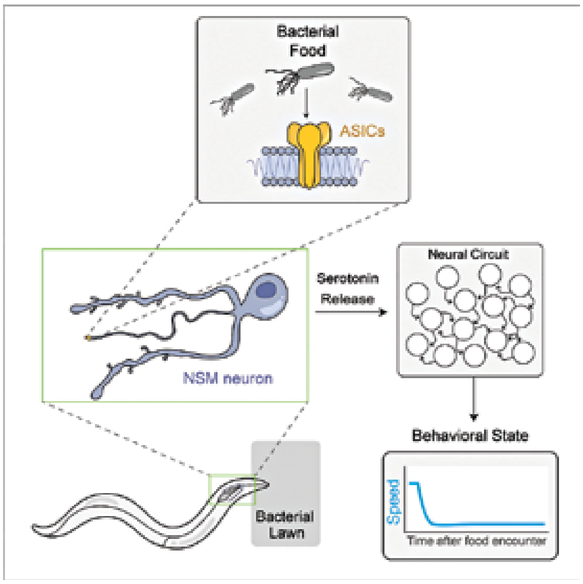
Once the neurites detect the bacteria, they signal the release of serotonin, which causes the organism to slow down its movement. The end result is that C. elegans stops to feed when there is a plentiful supply of food, and starts roaming around its environment when the food becomes scarce.

Now, the exact mechanism through which the ASICs detect the bacteria is not well understood. They might directly detect compounds secreted by the bacteria, or the ion channels may get activated by nearby receptors that detect the bacterial compounds. These ASICS exist in other animals including humans, and are associated with pain detection as well as taste. The functions of some other ASICs are still not known. The authors of the study suspect that the ASICs could have a general role to play when it comes to detecting bacteria in animal guts. It is unknown if the same mechanism for gut to brain signalling in the C. elegans also exists in humans as well. Understanding the exact mechanism in the worm may shed light on similar mechanisms in other animals, as well as humans.

In another study on the gut brain connection in the same organism, researchers from Brown University investigated another kind of behaviour. C. elegans has the capacity to distinguish between pathogenic bacteria and beneficial bacteria. This avoidance behaviour is trained or programmed into the worm, over a prolonged period of time, that may take up to four hours. The time gap allowed the researchers to suspect that the worm was understanding that its food source was noxious, because of signals sent from the intestine, where the pathogenic organisms accumulated. The researchers confirmed their suspicion by modifying cells found only in the intestine, and suppressing their capability to produce certain enzymes. When this was done, the worm took longer to identify a food source as toxic. Maintaining a balance in this regulation is important for survival, as if it is too strong, the worm would starve to death, and if it is too weak, then the worm will not be able to avoid the pathogens.

BUTTERFLIES IN THE STOMACH

For decades, researchers believed that mood changes caused stomach problems such as pain, constipation or diarrhoea. Recent research suggests that it is actually a two way connection, and the reverse can also occur. Irritation in the guts can cause changes in the mood, including anger and sadness. A larger than normal percentage of the population suffering from irritable bowel syndrome develop anxiety and depression. If you have ever felt nauseous before making a presentation, it is not something that you are imagining, and may actually be a physical response. Your mood may be changing the contractions of the digestive tract or aggravate inflammation. As the brain in the head and the brain in the guts are related to each other, the medical professionals who treat the problem in one can also help alleviate problems in the other. A gastroenterologist might prescribe antidepressants for example. Conversely, if a person is suffering from gastrointestinal problems, they may be encouraged to undergo therapies that reduce stress.



Credit: The Picower Institute

How the gut to brain signalling in the C. elegans works

There are trillions of bacteria in human bodies, and they also exist in the gut, in their own microbiome. In a simple organism such as the nematode, the bacteria in the guts elicit relatively simple responses, such as slowing down when there is plenty of food, or avoiding pathogenic food sources. In humans, the gut bacteria could interact with the body in more complex ways. The bacteria in the human gut regulate the immune response, and as such communicate with the human brain through the immune system.

There is increasing research to suggest that the interactions with the bacteria in the gut can have implications for a range of mental conditions, including Alzheimer's disease, Parkinson's disease and schizophrenia. In a groundbreaking study, researchers from the University of Chicago administered a long term course of antibiotics to lab mice. At the end of the course, they found that the composition of the bacteria in the mouse guts had changed considerably, and that there were fewer amyloid plaques forming in the brain, which is a build up of peptides or amino acid chains. The plaques are a sign of Alzheimer's. While there are implications for future treatments, this does not mean that antibiotics can cure Alzheimer's. At the very least, the finding shows that further research is required in the area, across multiple disciplines. Professor Sangram Sisodia who took the research in the unexpected direction noted, "We're exploring very new territory in how the gut influences brain health. This is an area that people who work with neurodegenerative diseases are going to be increasingly interested in, because it could have an influence down the road on treatments."